R6.4: the vertical margin, phase two

Round 6, 2026-10-19. Code `examples/r6_4_vertical_margin_2.rs` (native Rust). Raw records: `ladder_auto*.json`, `project.json`, `psi.json`, `modpoly.json`; fits: `analysis.json` / `analysis.txt`. Full write-up: `RESULTS.md`. Toy, synthetic instances only ($j = -3375$, $D = -7$, 61-bit p); no sub-rho or deployed-curve claim.

Question and hypothesis (preregistered)

What do ψ_ℓ factoring, $\Phi_{\ell(j,Y)}$ root finding and ascending edges cost, and does NTT move the generic exponent? Hypothesis: generic finding stays $\geq \ell^2$, the §17.3 crossover moves < 1 bit; ascending = descending. Falsifier: exponent CI below 2, or disjoint asc/desc intervals.

Method

Arms: (a) $\mathbb{F}_{p^k}$ ladder, `PolyMul::Auto`, $\ell = 37..863$, both directions; (a′) per-step counts measured exactly to $\ell = 2^{16}$ × exact step count (agrees with (a) to ≤ 0.44%); (b) ψ_ℓ + Cantor–Zassenhaus (crater) / gcd with $x^{p^k} - x$ (floor), $\ell \leq 131$; (c) library $\Phi_\ell \bmod p$ (uncounted, wall only) + counted evaluation and roots, $\ell \leq 103$. Unit Meq = M + W + 91 I, W stated at 1 M. 20/20 descending ladder rows reproduce R5.3.

Results (measured counts; (a′) projected)

series	range	exponent [95% CI]
(a) ladder desc	37..863	2.582 [2.544, 2.620]
(a) ladder asc	37..863	2.587 [2.547, 2.628]
(a′) projected Karatsuba	$\geq 2^{10}$	2.579 [2.553, 2.605]
(a′) projected Auto	$\geq 2^{10}$	2.088 [2.020, 2.155]
(a′) projected Auto	$\geq 2^{13}$	1.919 [1.724, 2.115]
(b) ψ + CZ desc	37..131	3.565 [3.302, 3.828]
(b) ψ + gcd asc	37..131	3.439 [3.087, 3.791]
(c) Φ eval+roots desc	11..103	1.884 [1.804, 1.964]
(c) Φ_ℓ mod p, wall s	20..103	4.600 [4.515, 4.684]

Paired asc/desc: ladder 1.001 [1.001, 1.002]; ψ 0.501 [0.457, 0.548]; Φ 0.360 [0.347, 0.373].

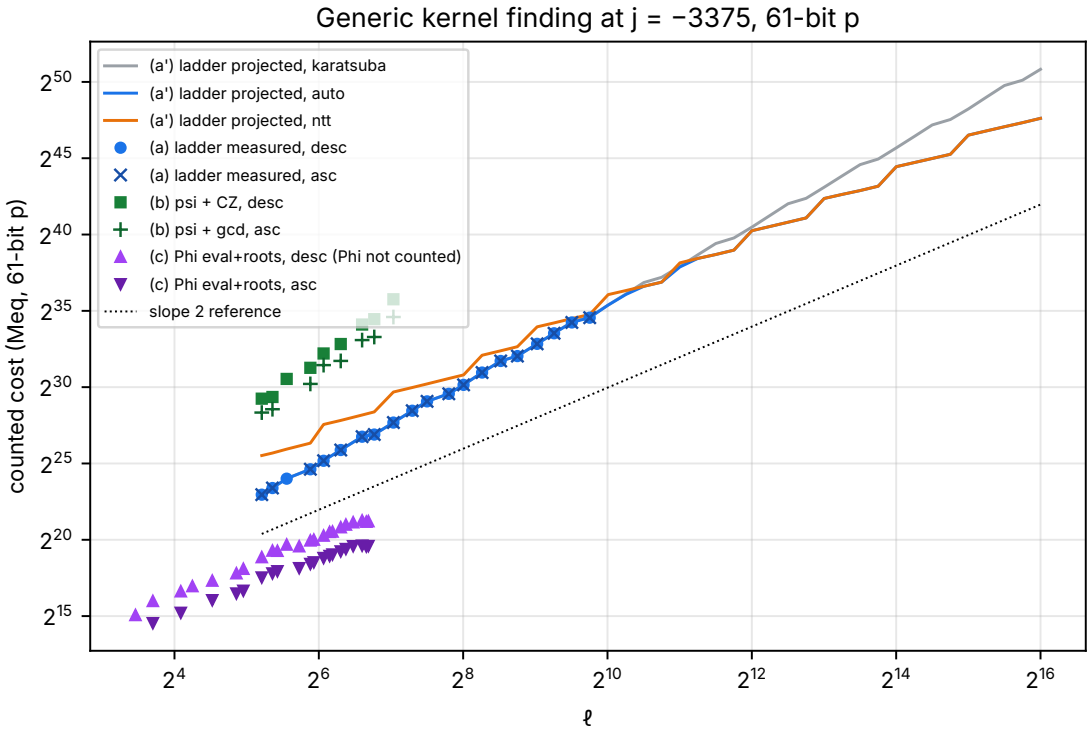


Figure 1: Generic kernel finding, all arms and both directions (measured; (a′) projected).

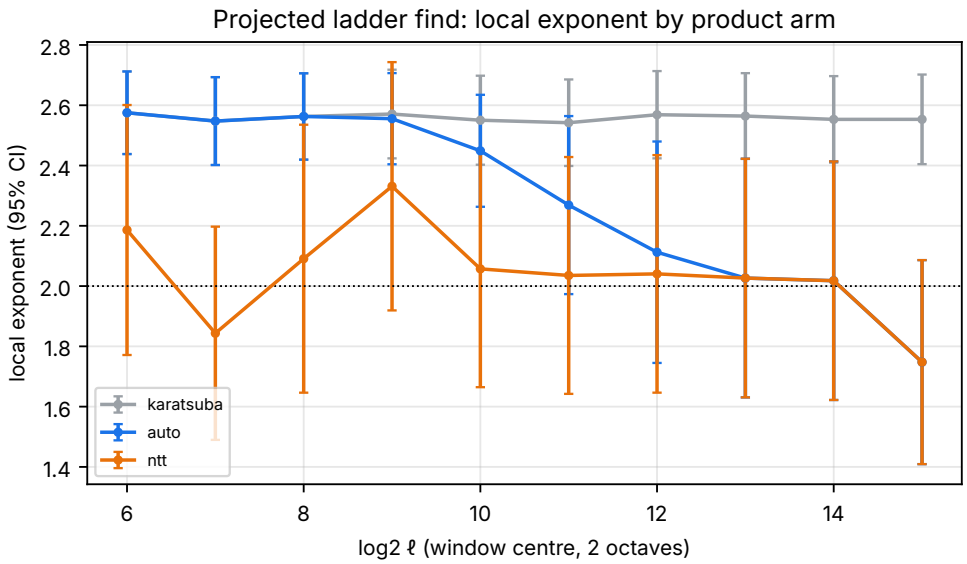


Figure 2: Local exponents of the projected ladder over 2-octave windows.

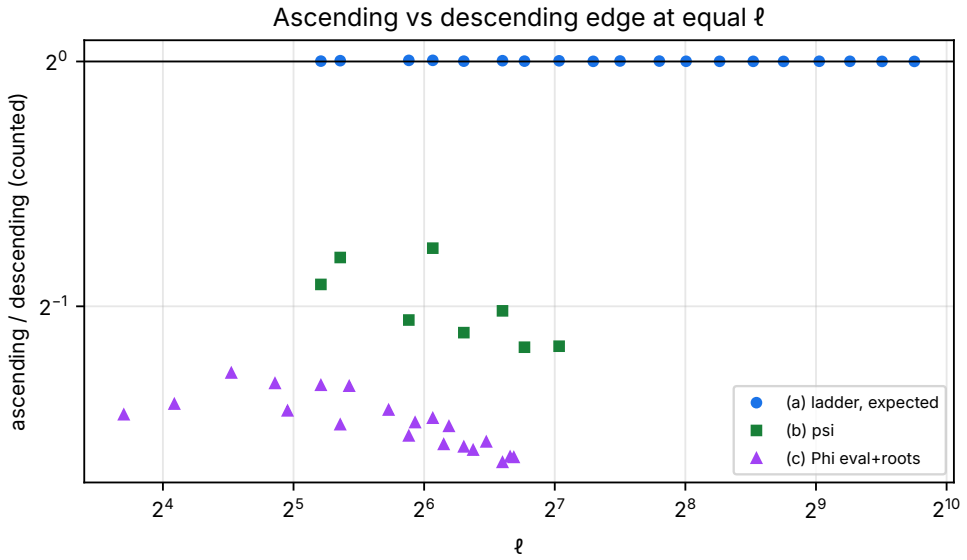


Figure 3: Ascending / descending counted cost at equal ℓ.

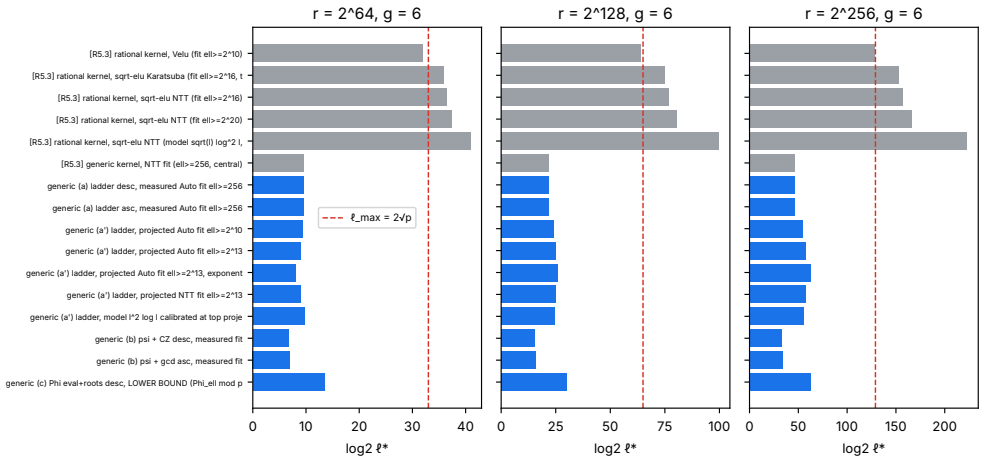


Figure 4: Derived (extrapolated) crossovers with rho.

Derived crossovers ($g = 6$, log2 ℓ^*)

model	$r = 2^{64}$	$r = 2^{128}$	$r = 2^{256}$
R5.3 / (a) measured fit	9.61	21.74	46.41
(a′) projected Auto, $\geq 2^{10}$	9.38	24.23	54.41
(a′) projected Auto, $\geq 2^{13}$	8.92	25.07	57.89
(b) ψ + CZ	6.69	15.38	33.05
(c) Φ, lower bound	13.52	29.98	63.42

Verdict

Exponent ≥ 2: **not falsified** (no CI below 2), but NTT moves it from 2.58 toward ≈ 2. Crossover moves < 1 bit: **refuted in projection** ($2^{21.7} \to 2^{24.2}..2^{25.1}$ at $r = 2^{128}$; more margin, not less). Asc = desc: **holds for the ladder**, **falsified for (b) and (c)** (floor factorisation structure); the cheapest route (a′) is direction-neutral. Whether a Sutherland-style Φ evaluation beats (a′) is open.

Limits

W = 1 M stated, not measured. (a′) is a projection beyond $\ell = 863$. Crossovers are extrapolations. Time caps: ψ at 131, Φ at 103; Φ_ℓ generation uncounted; one instance per ℓ.